

AI IN SECURITY CAMERAS: Real Change Or Hype?

Many brands have announced AI-powered security cameras. But, is it just a marketing hype or an actual game changer?

AKANKSHA SONDHI GAUR

Artificial Intelligence (AI) is rapidly reshaping security camera technology, equipping surveillance systems with advanced capabilities—from facial recognition to predictive analytics. AI is not only redefining how cameras operate but also expanding the scope of real-time monitoring. The global market, valued at US\$64.10 billion in 2023, is projected to reach US\$209.10 billion by 2033, growing at a CAGR of 12.55% over the forecast period. Rising adoption of AI and deep learning is driving this growth, significantly enhancing the effectiveness of surveillance.

Is AI in security cameras just a marketing hype?

Given the surge in AI-powered camera announcements, it is natural to question whether the systems deliver on their promises. To

summarise: while certain features remain in early development, the impact of AI in security is both tangible and significant. The ability to reduce false positives, automate surveillance tasks, and provide deeper insights into security data marks a major shift. However, as with any emerging technology, there are growing pains—AI models may occasionally underperform or require tuning. Still, the overall trend is clear: smarter and more efficient surveillance solutions are on the rise.

Edge vs cloud

The ‘edge versus cloud’ debate sits at the centre of AI-driven surveillance innovation. Some cameras process data directly on the device (edge AI), while others transmit footage to a central server for more complex analysis (cloud AI). Here is how the approaches differ:

- **Cloud AI.** This traditional architecture is still widely used. In this setup, AI-powered servers operate in the cloud. Data from the cameras is transmitted to these remote servers for processing. While this allows for more powerful and scalable machine learning models, it also introduces latency and demands high-bandwidth internet

connections. Cloud AI is particularly common in large-scale installations, where data from multiple cameras must be aggregated for pattern recognition and broader situational awareness.

- **Edge AI.** Thanks to advancements in electronics and camera hardware, it is now possible to analyse data and make real-time decisions directly on the device. A true edge AI system equips cameras with the computing power to capture, process, and respond in real-time, eliminating the need to transmit data externally. This significantly reduces reaction time, though it results in higher device costs due to more sophisticated hardware.

- **Server-on-Edge AI.** This emerging architecture blends the strengths of edge and cloud AI. In this model, the server is positioned close to the edge (i.e. near the cameras). This setup reduces latency and minimises bandwidth usage—critical for applications such as facial recognition, intruder alerts, and real-time anomaly detection. Server-on-Edge AI offers advantages such as faster processing, enhanced privacy, and greater reliability. The choice between architectures depends on the specific requirements of the security application.

What's changed? Learning capabilities of AI security cameras

The introduction of edge-based AI has notably reduced response times

